

1. What is the difference between Single quoted string and Double quoted string in Python?

Answer:

There is no major difference between single quotes (' ') and double quotes (" "). Both are used to create strings.

The main difference is that they are useful when the string contains the other type of quotation mark.

Example:

```
name = 'ABC'
```

```
message = "Hello, I'm ABC"
```

2. What is the difference between Immutable and Mutable objects?

Answer:

- **Mutable objects** can be changed after they are created.
- **Immutable objects** cannot be changed after they are created.

Examples:

- Mutable: list, set, dictionary
- Immutable: int, float, string, tuple

Example:

```
# Mutable
```

```
numbers = [1, 2, 3]
```

```
numbers[0] = 10
```

```
# Immutable
```

```
name = "ABC"
```

```
# name[0] = "A" # Not allowed
```

3. What is the difference between List and Tuple in Python?

Answer:

The main difference is **mutability**.

- **List** is mutable, so its elements can be changed.

- **Tuple** is immutable, so its elements cannot be changed.
- List uses square brackets [].
- Tuple uses parentheses ().

Example:

List

```
numbers = [1, 2, 3]
```

```
numbers[0] = 10
```

Tuple

```
numbers = (1, 2, 3)
```

```
# numbers[0] = 10 # Not allowed
```

4. What is the difference between a Set and List in terms of functionality and use cases?

Answer:

- A **List** stores multiple values in an ordered manner and allows duplicate values.
- A **Set** stores unique values and does not allow duplicates.
- Lists are useful when we need to maintain order.
- Sets are useful when we need unique values and fast membership checking.

Example:

List

```
numbers = [1, 2, 2, 3]
```

Set

```
numbers = {1, 2, 2, 3}
```

```
print(numbers)
```

```
# {1, 2, 3}
```

5. How does a Dictionary differ from a List in terms of data storage and retrieval?

Answer:

- A **List** stores data as an ordered collection of values and retrieves data using an **index**.
- A **Dictionary** stores data as **key-value pairs** and retrieves data using a **key**.
- Lists are useful when data has a specific order.
- Dictionaries are useful when we want to access data using meaningful keys.

Example:

List

```
student = ["ABC", 22, "Mumbai"]
```

```
print(student[0])
```

Dictionary

```
student = {"name": "ABC", "age": 22, "city": "Mumbai"}
```

```
print(student["name"])
```